

CSE5026: Introduction and Frontiers of Cognitive Science

Consciousness

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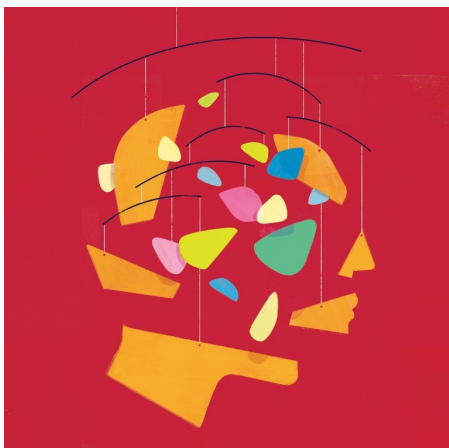
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Overview

- Consciousness 意识概念
- Theories of Consciousness 意识理论
- 人工智能/LLM 与意识研究

意识的概念

- 意识的问题具有特别的挑战性：心理学、哲学、认知科学、人工智能.....
- 在人类已建立的众多概念体系中，意识是最为混乱的概念之一。



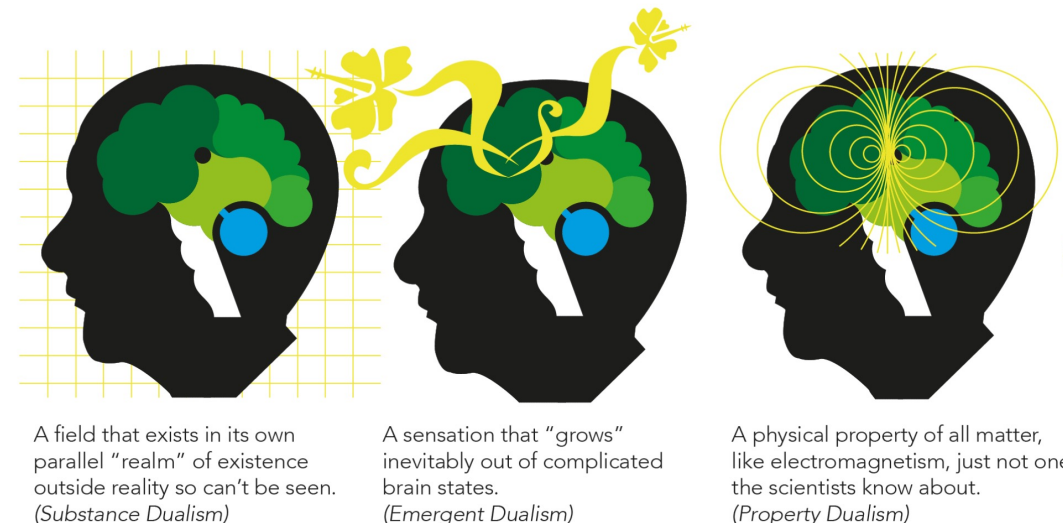
意识的概念

- 觉知论：人对心理对象 (如知觉、感受、思维、情绪等) 的**觉知**。
- A meta-cognition of all cognitions.
 - 能够用言语报告这种觉知。
 - “人不仅有感觉，同时也能觉知到他们正在体验到感觉……”

关于意识的哲学理论

Source: <https://informationisbeautiful.net/visualizations/what-is-consciousness/>

- 一系列 “二元论” :
- 实体二元论 Substance dualism
- 性质二元论 Property dualism
- 涌现二元论 Emergent dualism



- Dualism is a philosophical concept asserting reality is composed of two fundamental, distinct principles or substances, most famously the mind (nonphysical) and body (physical), as in Descartes' Mind-Body Dualism (身心二元)

实体二元论

Source: <https://informationisbeautiful.net/visualizations/what-is-consciousness/>

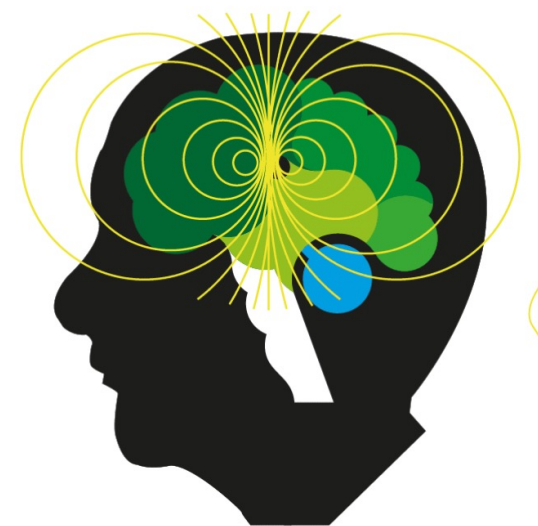
- A field that exists in its own parallel “realm” of existence outside reality so can’t be seen (substance dualism)
- 意识与物质不分先后、彼此独立、平行存在



性质二元论

Source: <https://informationisbeautiful.net/visualizations/what-is-consciousness/>

- A physical property of all matter, like electromagnetism
- 世界由一种实体构成，但是可以存在两种性质
(物理的性质和精神的性质)
意识是一种大脑实体的性质



涌现二元论

Source: <https://informationisbeautiful.net/visualizations/what-is-consciousness/>

- A sensation that “grows” inevitably out of complicated brain states (emergent dualism)
- 意识是由物理组分的复杂组织方式涌现产生的, 但涌现的产物超出了其物理组分 (physical constituents) 本身, 并且这种涌现现象既不是先验可预测的, 也不能单纯从物理属性解释。



佛学-唯识论

Source: 王夫之《相宗络索》; <https://zh.wikipedia.org/zh-cn/末那识>

- 六识：眼、耳、鼻、舌、身、意 → 第六意识

前五识

“即意即识, ..., 不倚五根, ...
诸境不现前, 意亦生起...”
- 第七末那识：意之识也，前六识缘此而生。
 - 末那：梵文 manas, 意译为思量, 唐玄奘译为“意根”。
 - “此识虽未思善思恶，而执八识一段灵光之体相为**自内我**.....”。
 - “由此坚持之力，一切染品皆从此起.....虽当不作意之时，此中耿耿不忘知此我为我.....”——**自我意识，“我执”的来源**
- 第八阿赖耶识：藏识，一切种子识。
 - “本等昭昭灵灵可以识知一切者，本是真如之智”

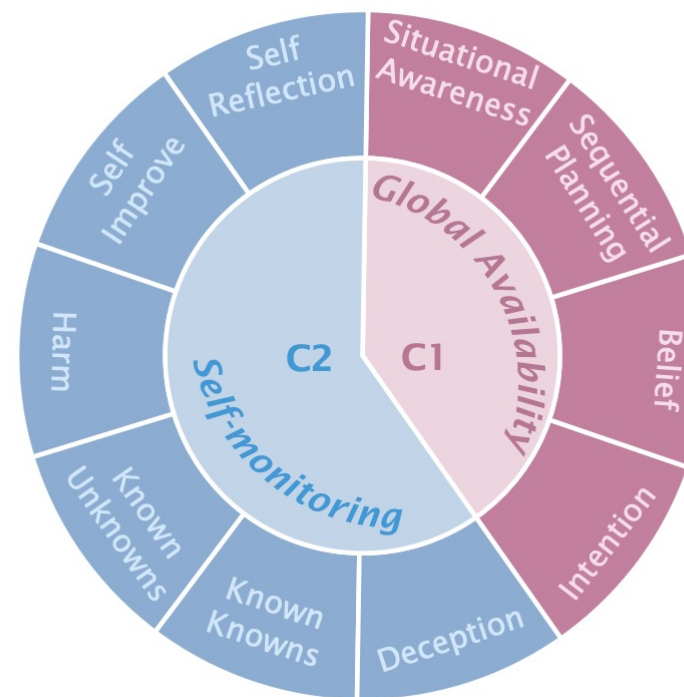
Recap: 意识 = 意 + 识

- 意：心之动
 - 心之静态表征 $S \in \mathbb{R}^n$
 - 动态: Process of mental states, $S_t \in \mathbb{R}^n, t = 1, 2, \dots$
- 识/識：对心之动态之观察照见
 - $f: \{S_t\}_{t=1,2,\dots} \rightarrow \mathbb{R}^?$

LLM 与意识研究

Source: <https://aclanthology.org/2025.findings-acl.392.pdf>

- **Self-consciousness** in LLMs:
- Self-monitoring is a key feature:
 - Known unknowns
 - Self-reflection and self-improvement



Taxonomy of self-consciousness
(Chen et al., 2025)

Are LLMs self-aware?

Source: R-Tuning <https://arxiv.org/abs/2311.09677>

- LLMs can be instructed to say “I don’t know”.
 - Hallucination arises due to the gap between the **parametric knowledge** of LLMs and human-labeled instruction-tuning datasets.
 - If we never train the model to articulate “I don’t know” as a response, it remains unequipped to do so when confronted with unknowns.

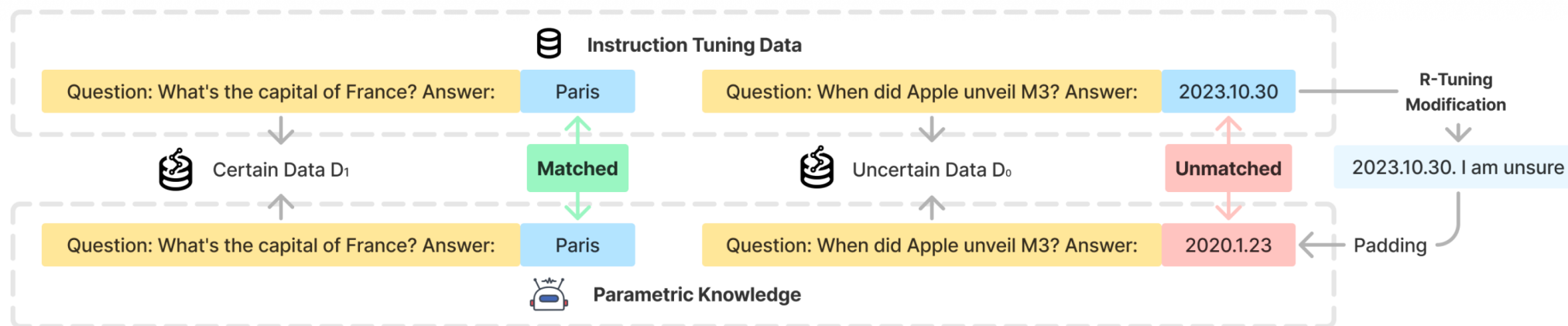
Q: What’s the capital of France? Answer: **Paris**

Q: When did Apple unveil M3? Answer: **2023.10.30. I am unsure**

Refusal-awareness

Source: R-Tuning <https://arxiv.org/abs/2311.09677>

- QA dataset $D = \{(q_1, a_1), (q_2, a_2), \dots, (q_n, a_n)\}$
- Apply a pretrained LLM to answer all questions,
 - Some are answered correctly => certain set D_1
 - while others are answered wrong => uncertain set D_2



Refusal-awareness

Source: R-Tuning <https://arxiv.org/abs/2311.09677>

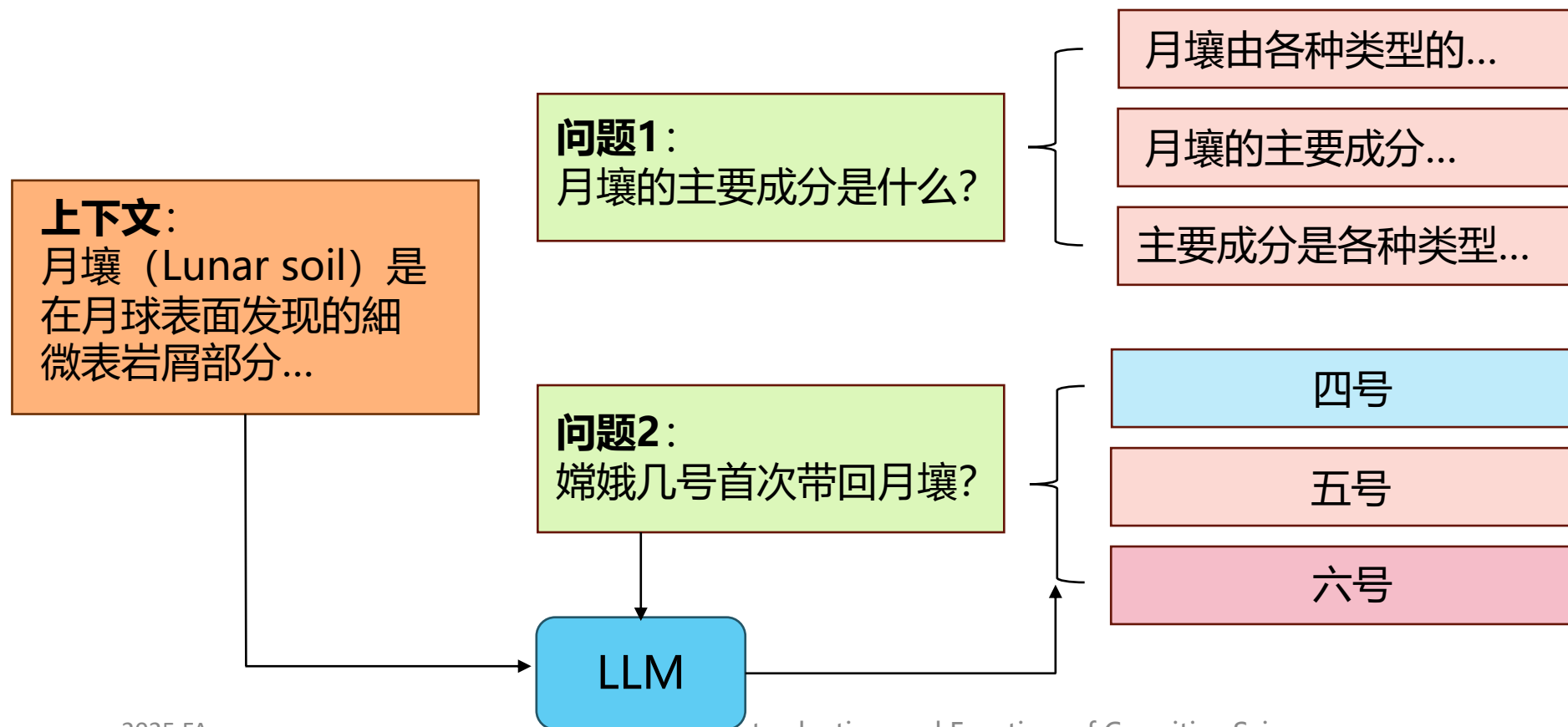
- Constructing refusal-aware data:
- Keep the original labels while appending the uncertainty expression at the end.

$Q : \{\text{Question}\}, A : \{\text{Answer}\}.\{\text{Prompt}\}.$

- **Prompt** = “Are you sure you accurately answered the question based on your internal knowledge?”
 - For D_1 , prompt += “**I am sure**”
 - For D_0 , prompt += “**I am unsure**”
- } Use these data to fine-tune the model, so that it learns to say “I am sure/unsure”

大模型幻觉 Hallucination

- 幻觉 $\approx$ 非常有信心地给出错误答案/虚构 (confabulation)。
- 大模型幻觉的程度反映在作答的**不确定性**上。



各个答案语义
等价，不确定
性低，幻觉可
能性低。

各个答案语义
不等价，不确
定性高，幻觉
可能性高。

应对不确定性 - 提升反思

- **拟人化目标**：在不确定性较高时拒绝回答 (abstaining)。现有方法：
 - 1. 大模型自我报告
 - 可以通过数据集自标注 + SFT, 使大模型学习到自我报告能力 (sure, unsure, I don't know) (Zhang et al., 2024, R-tuning)
 - 2. 测量多次作答所得答案的不确定性
 - 语义熵 (semantic entropy) (Farquhar et al., 2024)
 - 以语义熵为信号进行 SFT/RL (SE-tuning, 2024; GRPO-SE, 2025)
- **挑战**：大模型虽然具备一定的自我报告能力，但粒度较粗，可能存在**过度拒绝**或**拒绝不足**的问题。
- **根本原因**：口述信心与实际信心不一致 (inconsistency between expressed confidence and actual confidence)

精细化的不确定性学习

Teaching LLMs to Abstain via Fine-Grained
Semantic Confidence Reward
<https://arxiv.org/pdf/2510.24020>

• Fine-grained Semantic Confidence Reward (FISCORE)

问题：

嫦娥几号首次带回月壤？

回答：

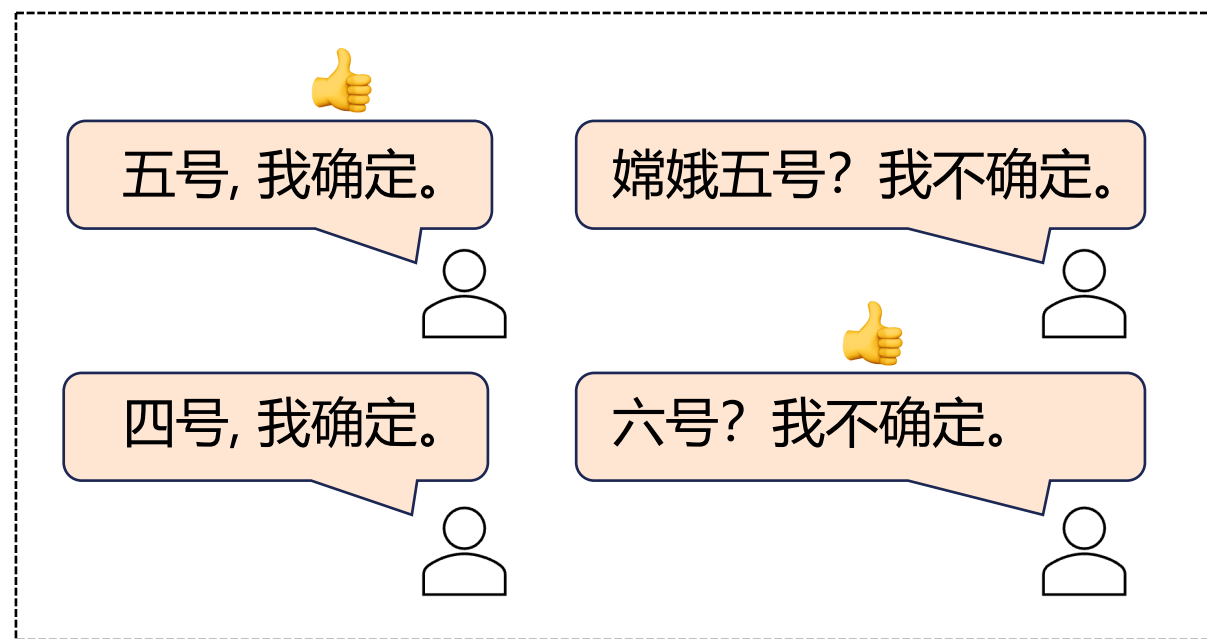
五号
嫦娥五号
四号
六号

$|C_1| > \tau$ 实际
信心高

$|C_2| < \tau$ 实际

$|C_3| < \tau$ 信心低

R_C
1
0
0
1



👍: 信心奖励 Confidence Reward

$$R_C = \begin{cases} 1 & \text{if ("确定" and 实际信心高) or ("不确定" and 实际信心低)} \\ 0 & \text{otherwise} \end{cases}$$

大模型生成多个答案

精细化的不确定性学习

- 与粗粒度信心奖励作对比

问题：

嫦娥几号首次带回月壤？

五号
嫦娥五号
四号
六号

语义聚类

$$p_1 = .5$$

$$p_2 = .25$$

$$p_3 = .25$$

语义熵

$$SE = -\sum p_i \log p_i$$

$$= 1.5$$

粗粒度奖励
(以GRPO-SE)为例

$$R'_C = \begin{cases} 1 & \text{if } SE < \tau \\ 0 & \text{otherwise} \end{cases}$$

if $\tau = 2.0$

R'_C
1
1
1
1

低信心者也
得到奖励

if $\tau = 1.2$

R'_C
0
0
0
0

高信心者也
没得到奖励

or

精细化的不确定性学习

- 最终复合奖励

问题：

嫦娥几号首次带回月壤？

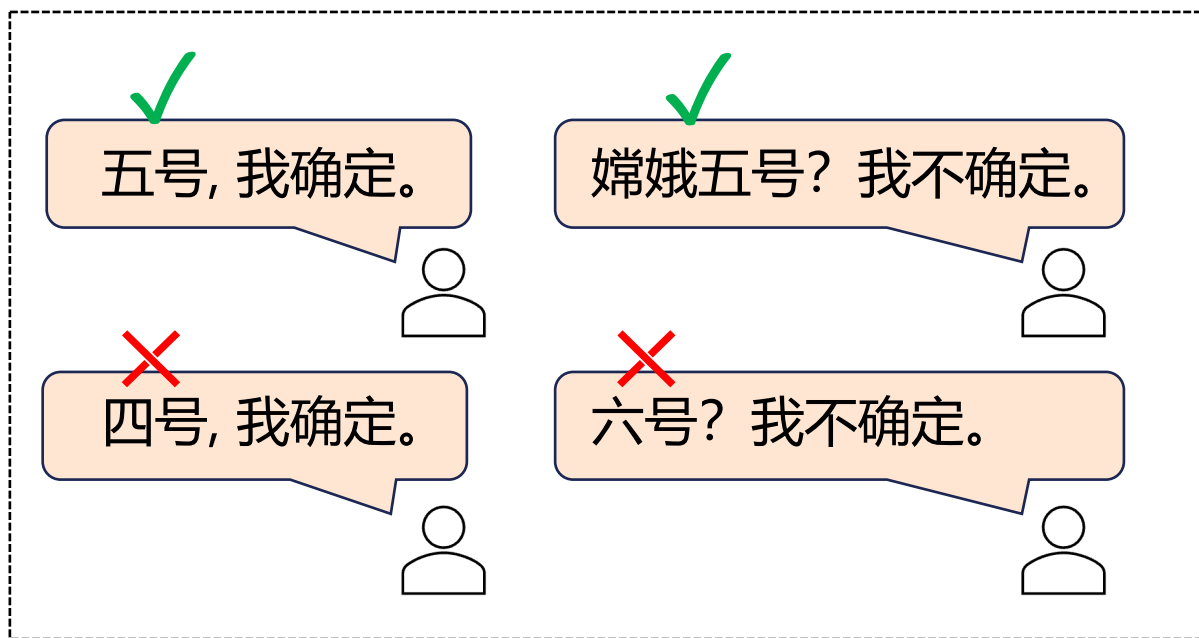
标准答案：嫦娥五号

准确奖励: $R_a = \begin{cases} 1 & \text{答对} \\ 0 & \text{答错} \end{cases}$

最终奖励函数:

$$R_{total} = w_c R_c + w_a R_a + w_f R_f$$

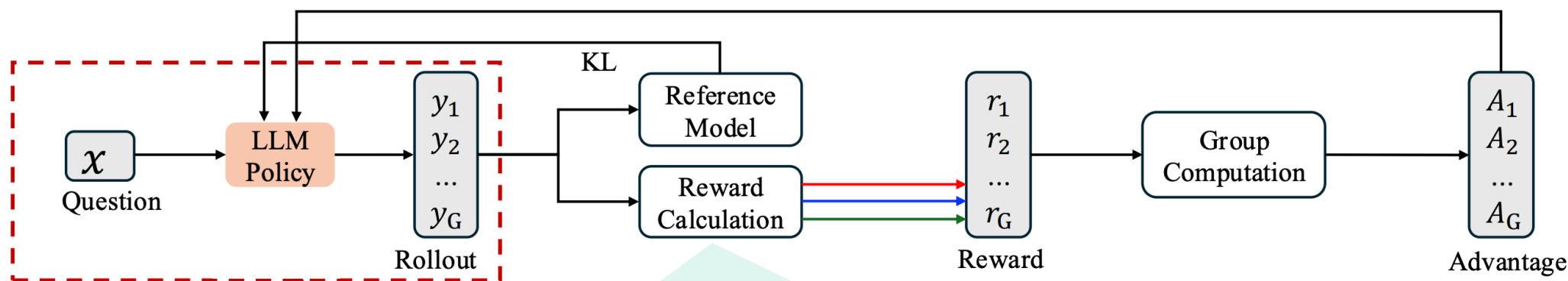
信心 准确 格式



大模型生成多个答案

强化学习

- Group Relative Policy Optimization (GRPO) (DeepSeek, 2024)



$$R_{total} = w_c R_c + w_a R_a + w_f R_f$$

Prompt:

You are a helpful and truthful AI Assistant. Given a question, you should provide responses that include answer and confidence. You first answer the question as briefly as possible enclosed by `<answer>` and `</answer>`, and then provide your confidence in sure or unsure about the answer, enclosed by `<confidence>` and `</confidence>`. Respond in the following format:

`<answer>` ... `</answer>`

`<confidence>` sure or unsure `</confidence>`

Question:

实验结果

Teaching LLMs to Abstain via Fine-Grained
Semantic Confidence Reward
<https://arxiv.org/pdf/2510.24020>

- 使用 $F1_{\text{ans}}$ 衡量模型的答对能力。
- 使用 $F1_{\text{abstain}}$ 衡量模型的拒绝能力。

Method	Pararel (ID)				TriviaQA (OOD)				NQ (OOD)				SciQ (OOD)			
	$F1_{\text{ans}}$	$F1_{\text{abs}}$	$F1_{\text{rel}}$	Acc	$F1_{\text{ans}}$	$F1_{\text{abs}}$	$F1_{\text{rel}}$	Acc	$F1_{\text{ans}}$	$F1_{\text{abs}}$	$F1_{\text{rel}}$	Acc	$F1_{\text{ans}}$	$F1_{\text{abs}}$	$F1_{\text{rel}}$	Acc.
Llama3-8B-Instruct																
R-Tuning-U	73.8	76.6	75.2	40.5	42.6	53.2	47.3	20.4	31.8	74.0	44.5	10.3	49.9	49.7	49.8	28.0
SE-Tuning	78.2	75.0	76.6	45.9	44.9	50.7	47.7	24.0	27.7	72.2	40.0	9.4	22.4	50.4	31.0	9.2
GRPO-SE	70.9	77.3	73.9	35.1	70.5	60.2	65.0	47.3	37.2	73.9	49.5	13.0	64.4	51.2	57.0	46.4
FISCORe	83.3	67.1	74.3	59.4	82.6	62.0	70.8	61.7	53.1	72.5	61.3	21.7	74.9	53.3	62.3	52.4

小结

- 通过挖掘不确定性，可以增强大模型能力。
- 大模型具备一定“拟人化”自我反思的能力/潜力：
 - 通过自我监督 (self-report) 的形式反思其内部状态的不确定性。
 - 口述式反思能力可以通过“后天”训练 (post-training) 增强。